AIM: Containerize an application using docker
LO:04
LO:
THEORY:
1. What is Docker?
Docker is an open-source platform that allows developers to build, package, and run applications inside containers. It provides tools to create images (blueprints of applications), manage containers (running instances of those images), and share them through registries like Docker Hub. By using Docker, applications can be deployed consistently across different environments, eliminating compatibility issues and simplifying development, testing, and deployment.
2. What is Containerization?
Containerization is the process of bundling an application together with all its dependencies, libraries, and configuration files into a single lightweight unit called a container. Unlike virtual machines, containers don't require a full operating system for each instance; they share the host OS kernel while remaining isolated. This makes them faster, more efficient, and portable, ensuring that applications run reliably across different systems and environments
3. What is the difference between Image and Containerization?
Docker Image: Docker image is a blueprint or template that contains everything needed to run an application — the code, dependencies, libraries, and configuration. Images are read-only and act as the starting point for creating containers. You can think of an image as a snapshot of an application environment.
Containerization: Containerization is the method of packaging applications into isolated units called containers. It ensures that the application runs consistently across different environments by bundling the app with its dependencies. Containers are created from images and provide the actual runtime environment where the application executes.
Commands:
1. docker - - version
2. docker search ubuntu

3. docker search ubuntu - -filter stars=100
4.

```
C:\Users\15L>docker --version
Docker version 25.0.3, build 4debf41
```

```
C:\Users\15L>docker search ubuntu
NAME                              DESCRIPTION                                     STARS     OFFICIAL
ubuntu                            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                      Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                      BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                      Apache Kafka, a distributed event streaming …   62
ubuntu/apache2                    Apache, a secure & extensible open-source HT…   104
ubuntu/zookeeper                  ZooKeeper maintains configuration informatio…   15
ubuntu/prometheus                 Prometheus is a systems and service monitori…   80
ubuntu/jre                        Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet              Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                     A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                      MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                   PostgreSQL is an open source object-relation…   45
ubuntu/redis                      Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-runtime             Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps                Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                    Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                  Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                        Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                   Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                      Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                       Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                    Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector    This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager               Ubuntu Rock for Alertmanager. It handles ale…   1
```

```
C:\Users\15L>docker search ubuntu --filter stars=100
NAME              DESCRIPTION                                     STARS     OFFICIAL
ubuntu            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx      Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9      BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/apache2    Apache, a secure & extensible open-source HT…   104
```

```
C:\Users\15L>docker run ubuntu /bin/bash
```

```
C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

```
C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
ed819469700f: Pull complete
a3679419df18: Pull complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

```
C:\Users\15L>docker run -it ubuntu /bin/bash
root@4b486d8501d7:/#
```

```
root@4b486d8501d7:/# exit
exit
```

```
C:\Users\15L>docker ps -a
CONTAINER ID   IMAGE                              COMMAND                  CREATED          STATUS
 PORTS                    NAMES
4b486d8501d7   ubuntu                             "/bin/bash"              16 minutes ago   Exited (127) 5 minutes ago
                        silly_keldysh
b15df8ca655b   ca2b0f26964c                       "/bin/bash"              19 minutes ago   Exited (0) 17 minutes ago
                        sleepy_yonath
79d8deb2f28e   nginx                              "/docker-entrypoint.…"   8 months ago     Created
                        ecstatic_satoshi
065e20c61331   nginx                              "/docker-entrypoint.…"   8 months ago     Exited (255) 20 minutes ago
 0.0.0.0:8000->80/tcp   keen_morse
4e2f1b0ebad1   nginx                              "/docker-entrypoint.…"   8 months ago     Exited (0) 8 months ago
                        angry_torvalds
578a8f30694f   nginx                              "/docker-entrypoint.…"   8 months ago     Created
                        relaxed_jepsen
4b2344abd78a   mysql:latest                       "docker-entrypoint.s…"   2 years ago      Exited (1) 2 years ago
                        practical_spence
66c4141a85c5   mysql:8.3.0                        "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago
                        mydb
0a805a3dc56c   ca2b0f26964c                       "/bin/bash"              2 years ago      Exited (0) 2 years ago
                        vibrant_lamarr
7fec2031b676   0e7372b5f723                       "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago
                        trusting_newton
df0f11f819ff   docker/welcome-to-docker:latest    "/docker-entrypoint.…"   2 years ago      Exited (137) 2 years ago
                        welcome-to-docker
```

```
C:\Users\15L>docker ps --last 1
CONTAINER ID   IMAGE     COMMAND       CREATED          STATUS                       PORTS     NAMES
4b486d8501d7   ubuntu    "/bin/bash"   19 minutes ago   Exited (127) 10 minutes ago            myContainer
```

```
C:\Users\15L>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
062e450697fa: Pull complete
82454cdbf456: Pull complete
3c7ab7949321: Pull complete
cacfcdd01f30: Pull complete
b6698f04e005: Pull complete
2bedaf25031a: Pull complete
d26f27cc8c41: Pull complete
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

```
C:\Users\15L>docker images
REPOSITORY                TAG       IMAGE ID       CREATED        SIZE
nginx                     latest    4e5db4761e0f   4 days ago     161MB
ubuntu                    latest    de7345b16e94   6 days ago     100MB
nginx                     <none>    9d0e6f6199dc   8 months ago   152MB
welcome-to-docker         latest    def189b16f2a   2 years ago    226MB
<none>                    <none>    0e7372b5f723   2 years ago    226MB
ubuntu                    <none>    ca2b0f26964c   2 years ago    77.9MB
mysql                     8.3.0     019814493c7a   2 years ago    632MB
mysql                     latest    019814493c7a   2 years ago    632MB
redis                     latest    170a1e90f843   2 years ago    138MB
docker/welcome-to-docker  latest    c1f619b6477e   2 years ago    18.6MB
nginx/docker-extension    0.0.3     41d3d0d7d940   3 years ago    7.53MB
```

```
C:\Users\15L>docker info
Client:
 Version:    25.0.3
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.12.1-desktop.4
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v2.24.5-desktop.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container. (Docker Inc.)
    Version:  0.0.24
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  dev: Docker Dev Environments (Docker Inc.)
    Version:  v0.1.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-dev.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.21
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
  feedback: Provide feedback, right in your terminal! (Docker Inc.)
    Version:  v1.0.4
    Path:     C:\Program Files\Docker\cli-plugins\docker-feedback.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.0.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-init.exe
  sbom: View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc.)
    Version:  0.6.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-sbom.exe
```

```
C:\Users\15L>docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
eb601c099809   bridge    bridge    local
0a2f5b91dd47   host      host      local
0dc9050409db   none      null      local
```

```
C:\Users\15L>docker run nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/20 10:35:40 [notice] 1#1: using the "epoll" event method
2026/07/20 10:35:40 [notice] 1#1: nginx/1.31.3
2026/07/20 10:35:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/20 10:35:40 [notice] 1#1: OS: Linux 5.15.133.1-microsoft-standard-WSL2
2026/07/20 10:35:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/20 10:35:40 [notice] 1#1: start worker processes
2026/07/20 10:35:40 [notice] 1#1: start worker process 29
2026/07/20 10:35:40 [notice] 1#1: start worker process 30
2026/07/20 10:35:40 [notice] 1#1: start worker process 31
2026/07/20 10:35:40 [notice] 1#1: start worker process 32
2026/07/20 10:35:40 [notice] 1#1: start worker process 33
2026/07/20 10:35:40 [notice] 1#1: start worker process 34
2026/07/20 10:35:40 [notice] 1#1: start worker process 35
2026/07/20 10:35:40 [notice] 1#1: start worker process 36
2026/07/20 10:35:40 [notice] 1#1: start worker process 37
2026/07/20 10:35:40 [notice] 1#1: start worker process 38
2026/07/20 10:35:40 [notice] 1#1: start worker process 39
2026/07/20 10:35:40 [notice] 1#1: start worker process 40
```